

9.1 Part 1: Practical biology skills

Students will further develop their practical skills, whichever approach (context or concept) has been taken. Students will carry out the recommended core practicals and other practical investigations, which will require them to work safely, produce valid results and present data in the most appropriate format.

Students will carry out practical work during the GCE Biology AS course, which will be verified by the teacher using the criteria below and submitted to Edexcel using a verification of practical skills record.

The teacher will verify students' ability to:

- a Use apparatus skilfully and safely
 - i Apparatus and materials are handled correctly and safely and manipulative techniques are used in an appropriate and safe manner.
 - ii The practical work is carried out in an organised, methodical and safe manner, with due consideration of the wellbeing of living organisms and the environment.
- b Produce and record reliable and valid results
 - i Measurements and observations are made with precision and recorded in a structured manner; variables are identified and the validity and reliability of results are justified.
 - ii Possible systematic errors and random errors in generating results are identified and explained.
- c Present and analyse data
 - i Use appropriate methods to analyse results, present data and identify trends, patterns and/or observations.
 - ii Any apparent anomalies and inconsistencies are described, the methodology is evaluated and suggestions are made to improve or further the work of the investigation.

Edexcel reserves the right to:

- make it a requirement that centre staff undertake appropriate training to ensure the correct application of the verification process of practical skills
- arrange a centre visit to inspect procedures for the verification of practical skills.

Other practical-related skills, including analysis and evaluation of data may be assessed in the externally assessed components.